Universal Print Deployment & Print-Server Retirement

Implementation Solution Guide

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Executive Summary — Universal Print Deployment & Print-Server Retirement

Print servers are a quiet liability: a box (or several) you patch, back up, and pray never falls over on a Monday morning, plus driver sprawl and GPO printer mappings that only work on the corporate LAN. **Microsoft Universal Print** moves printing to a Microsoft-hosted cloud service — printers are shared from Entra ID, deployed to users and devices through Intune, and reachable wherever people work — so the print servers can be decommissioned entirely.

This is a complete, deploy-it-today package for migrating to Universal Print and retiring print servers, written for an administrator with no prior Universal Print experience. It uses a coexistence-then-cutover approach: connect the existing printers, prove cloud printing on a pilot, then retire the old queues and servers.

What you get

- A plain-language **Solution Overview** of how Universal Print works — the connector model, Entra ID printer sharing, Intune-driven printer deployment, and the path to retiring print servers.
- A **Requirements Guide** with verify / implement-if-missing for every prerequisite.
- An **Implementation Guide** of precise, click-by-click steps in the required order.
- Five **PowerShell scripts** for print-environment discovery, Universal Print readiness and print validation, a printer/deployment inventory report, safe remediation of failed deployments, and a printer-deployment group builder.
- **Testing** and **Troubleshooting** guides for confident rollout, plus field-proven **Best Practices.**

What this delivers

Cloud-delivered printing for every user — printers registered through a Universal Print connector, shared from Entra ID, and deployed by group through Intune so they appear automatically on managed devices, reachable on or off the corporate network. Once the pilot validates printing, the old print queues and servers are retired in a controlled cutover.

The repeatable deployment model

Assess → Configure → Pilot → Validate → Expand — with a pilot gate before any server retirement.

Outcome

Printers shared from Entra ID and deployed through Intune, reachable from anywhere, with the print servers decommissioned — proven on a pilot before you scale, and deployable by a single administrator.

1. Solution Overview — Universal Print Deployment & Print-Server Retirement

What Universal Print is

Microsoft Universal Print is a cloud-based print service hosted in Microsoft 365. Instead of an on-premises print server hosting queues and drivers, printers are registered in the Universal Print service, **shared from Entra ID**, and **deployed to users and devices through Intune**. Managed Windows devices receive the printers automatically and print over HTTPS to the cloud service, which routes the job to the printer — on or off the corporate network. The goal of this solution is to get there in an orderly way and then **retire the print servers**.

How it works

1. A printer reaches Universal Print one of two ways: it is **Universal Print-ready** (registers itself directly with the service), or it is a normal printer reached through a **Universal Print connector** — a small Windows service you install on a host that already sees those printers.
1. Once registered, an administrator **shares** the printer in the Universal Print portal. The share becomes an object in Entra ID that you can grant members and deploy.
1. **Intune printer deployment** assigns shared printers to Entra ID **groups**. Devices (or users) in a targeted group receive the printer automatically — no driver staging, no GPO mapping.
1. A user prints; the device sends the job to Universal Print over HTTPS; the service hands it to the printer directly (Universal Print-ready) or back through the connector to the physical printer.
1. Because routing is cloud-side, printing works the same **on the LAN, over VPN, or from home** — as long as the device and the connector host can reach the service.

Why retire the print server

The connector is not a permanent replacement for a print server — it is the **bridge** that lets existing printers participate in Universal Print during migration. Once printers are registered, shared, deployed by Intune, and validated on a pilot, the old print queues and the server hosting them no longer carry production traffic and can be decommissioned. That removes a patched, backed-up box, ends driver sprawl, and eliminates LAN-only printing — the whole reason for the migration.

What you can control

- Which printers are registered (connector-attached vs. Universal Print-ready) and their **share names.**
- Who can use each printer — membership of the printer's Entra ID share and the deployment group.
- How printers reach devices — **Intune printer deployment** targeted by group, per device or per user.
- Coexistence: run the old queues and Universal Print **side by side** during the pilot, then cut over.

Benefits

- **No print servers** — decommission the boxes you patch, back up, and worry about.
- **Print from anywhere** — cloud routing works on or off the corporate network.

• **Assign once, repeat everywhere** — deploy by group; new users in the group get the printer automatically.

• **Less driver sprawl** — devices use the in-box Universal Print client, not per-model drivers and GPO scripts.

• **Identity-based access** — printer access is governed through Entra ID groups.

Limitations (be honest with buyers)

• Non-Universal-Print-ready printers need a **connector host** that stays online to relay jobs; that host must be reachable for those printers to print.

• Each connector has a printer-count and throughput ceiling; large estates may need more than one connector.

• Advanced finishing/print-accounting features depend on printer and connector support; verify per model.

• Devices and the connector host need reliable outbound HTTPS to the Universal Print endpoints.

• Universal Print requires eligible Microsoft 365 licensing with print volume allotment (see requirements).

The repeatable deployment model

> *Register existing printers via the connector → share them in Entra ID → deploy to groups via Intune → validate on a pilot (on and off network) → retire the old queues/servers → expand. Pattern: Assess → Configure → Pilot → Validate → Expand, with a pilot gate before server retirement.*

2. Requirements Guide — Universal Print Deployment & Print-Server Retirement

Verify **every** requirement before configuring. Each entry: Description · Why · Verify · Implement-if-missing. Category-wide prerequisites are in `global-requirements-infrastructure.md`.

> *Naming convention used throughout this solution (reuse verbatim):* `<Purpose> – <Scope> – <Stage>`*. The objects you will create are: - Connector host name:* `UniversalPrint – Connector – Site1` *- Registered/shared printer name:* `UniversalPrint – <Model/Location> – Shared` *(e.g.* `UniversalPrint – HQ-Floor2-MFP – Shared`*) - Printer-deployment group (pilot):* `UniversalPrint – Printers – Pilot` *- Printer-deployment group (broad):* `UniversalPrint – Printers – Broad`

2.1 Universal Print licensing

Description: Eligible Microsoft 365 licensing that includes Universal Print and print-volume allotment. Why: Universal Print is a licensed Microsoft 365 service; registered printers consume print-job allotment. Verify: Microsoft 365 admin center → **Billing** → **Licenses**; the Universal Print portal loads at `https://portal.azure.com` → search `Universal Print`. Implement: Assign a qualifying license (e.g. Microsoft 365 E3/E5, or a Universal Print add-on/Windows Enterprise plan that includes it); confirm

print-volume allotment.

2.2 Entra ID and Intune (deployment plane)

Description: An Entra ID tenant with Intune, to share printers and deploy them by group. Why: Printer sharing is an Entra ID object; Intune printer deployment targets Entra ID groups. Verify: `https://entra.microsoft.com` and `https://intune.microsoft.com` both load. Implement: See `global-requirements-infrastructure.md` G1–G3.

2.3 Connector host (for existing printers)

Description: A Windows host (Windows 10/11 or Windows Server) that already reaches your existing printers, to run the Universal Print connector. Why: Non-Universal-Print-ready printers reach the service only through the connector; the host must stay online. Verify: Run `UPRT-Validation.ps1`; confirm the candidate host can ping/print to the target printers and has outbound HTTPS. Implement: Choose a reliable always-on host with line-of-sight to the printers; you will install the connector in the Implementation Guide.

2.4 Existing print inventory (servers, queues, drivers)

Description: A known inventory of current print servers, shared queues, and drivers to migrate. Why: You cannot retire what you have not accounted for; the inventory drives the registration list. Verify: Run `UPRT-Discovery.ps1` against each print server; run `UPRT-Report.ps1` for a consolidated list. Implement: Capture the printer list, ports, and drivers; mark which printers are Universal Print-ready vs. connector-attached.

2.5 Network / firewall reachability

Description: Outbound HTTPS from devices and the connector host to the Universal Print endpoints. Why: Devices submit jobs and the connector relays them to the service over HTTPS; blocked endpoints stall printing. Verify: From a device and the connector host, the Universal Print sign-in and service endpoints are reachable (run `UPRT-Validation.ps1`). Implement: Allow Microsoft's published Universal Print / Microsoft 365 URLs and IP ranges through the proxy/firewall.

2.6 Universal Print Administrator role

Description: The operator holds the **Printer Administrator** (or Global Administrator) role for Universal Print. Why: Registering, sharing, and managing printers requires the Universal Print admin role. Verify: Universal Print portal → **Settings** → **role** shows Printer Administrator; or Entra → **Roles and administrators**. Implement: Assign **Printer Administrator** to the operator in Entra ID.

2.7 Pilot and broad deployment groups

Description: Entra ID security groups for the printer-deployment pilot and broad populations. Why: Intune deploys printers by group; the pilot group is the gate before retirement. Verify: Entra → **Groups**; `UniversalPrint – Printers – Pilot` and `UniversalPrint – Printers – Broad` exist. Implement: Create them (Implementation Step 5 builds them, or use `UPRT-Set-PrinterDeploymentTargeting.ps1`).

Checklist

- Universal Print licensing assigned (with print-volume allotment)
- Entra ID + Intune available (G1–G3)
- Connector host selected and reachable — `UPRT-Validation.ps1`
- Existing print inventory captured — `UPRT-Discovery.ps1` / `UPRT-Report.ps1`
- Network allows Universal Print endpoints — `UPRT-Validation.ps1`
- Printer Administrator role assigned
- `UniversalPrint – Printers – Pilot` and `UniversalPrint – Printers – Broad` groups created

3. Implementation Guide — Universal Print Deployment & Print-Server Retirement

Follow these steps **in order**. You will work in the **Universal Print portal** (open `https://portal.azure.com` and search `Universal Print`, or use the Microsoft Intune admin center), on a **Windows connector host**, and in the **Microsoft Intune admin center** (`https://intune.microsoft.com`).

> *Model: register existing printers through a connector, share them in Entra ID, deploy by group via Intune, validate on a pilot on and off network, then retire the old queues and servers. Reuse the names from the Requirements Guide verbatim.*

Step 0 — Assess the current print environment

1. Run `UPRT-Discovery.ps1` on each print server to list queues, ports, and drivers.
2. Run `UPRT-Report.ps1` for a consolidated printer inventory; mark each printer as Universal Print-ready or connector-attached.

Step 1 — Enable and confirm Universal Print licensing

1. In the Universal Print portal (`https://portal.azure.com` → search `Universal Print`), open **Overview** and confirm the service is active and licensed.
2. Microsoft 365 admin center → **Billing** → **Licenses** — confirm the Universal Print allotment and that operators hold the **Printer Administrator** role.

Step 2 — Install the Universal Print connector and register existing printers

1. On the chosen connector host, name the machine `UniversalPrint – Connector – Site1` (record this; it identifies the connector in the portal).
2. Download the **Universal Print connector** from the Universal Print portal (**Connectors** → **Add**, follow the download link) and install it on the host.

3. Launch the connector, sign in with the **Printer Administrator** account, and register the host. The connector enumerates the printers the host can already see.

4. Select the existing printers to bring into Universal Print and **Register** them. They now appear in the portal under **Printers**.

5. (Universal Print-ready printers register themselves directly from the device's own settings — no connector needed; confirm they appear under **Printers**.)

Step 3 — Share the printers in Entra ID

1. Universal Print portal → **Printers** → select a registered printer → **Share**.

2. Set the **Share name** to `UniversalPrint – <Model/Location> – Shared` (e.g. `UniversalPrint – HQ-Floor2-MFP – Shared`). Reuse this convention for every printer.

3. Under **Members**, you can grant users/groups here, but for managed devices prefer Intune deployment (Step 5). **Share**.

4. Repeat for each printer being migrated.

Step 4 — Verify a shared printer manually (pre-pilot smoke test)

1. On your admin device, **Settings** → **Bluetooth & devices** → **Printers & scanners** → **Add device** → **Search Universal Print for printers**, add the shared printer, and print a test page.

2. Confirm the job completes through the connector (or directly for Universal Print-ready). If this fails, fix it before deploying broadly — run `UPRT-Validation.ps1`.

Step 5 — Build the printer-deployment groups

1. Entra → **Groups** → **New group** → Security → `UniversalPrint – Printers – Pilot` (a few representative devices/users). **Create**.

2. Create `UniversalPrint – Printers – Broad` (the rest of the population). **Create**.

3. Alternatively, run `UPRT-Set-PrinterDeploymentTargeting.ps1` to create these groups and stage their membership.

Step 6 — Deploy printers to the pilot via Intune

1. Intune → **Devices** → **Manage devices** → **Configuration** → **Universal Print printer deployment** (or **Devices** → **Windows** → **Universal Print printer deployment**) → **Add / Create**.

2. Select the shared printer (e.g. `UniversalPrint – HQ-Floor2-MFP – Shared`).

3. **Assignments** → **Add groups** → `UniversalPrint – Printers – Pilot`. **Create**.

4. Repeat for each pilot printer. Targeted devices receive the printer automatically on next sync.

Step 7 — Validate the pilot (on and off network)

1. On a pilot device synced to policy, confirm the deployed printer appears in **Printers & scanners** without manual add.

2. Print a test page **on the corporate LAN**, then **off network** (VPN off / home / hotspot) and confirm both complete.

3. Run `UPRT-Validation.ps1` (readiness + a test print) and `UPRT-Report.ps1` (deployment coverage). Complete the Testing Guide.

Step 8 — Pilot gate, then expand to broad

1. Only after the pilot passes (Testing Guide gate) add the broad population: edit each printer deployment → **Assignments** → **Add groups** → `UniversalPrint – Printers – Broad`.

2. Re-validate coverage with `UPRT-Report.ps1`; remediate any failed deployments with `UPRT-Remediation.ps1` (default `-WhatIf`).

Step 9 — Retire the old print queues and servers (coexistence → cutover)

1. During coexistence, the old queues and Universal Print run side by side. Watch the old print server's job logs; as Universal Print coverage grows, traffic on the old queues should fall to near zero.

2. When a printer's users are fully on Universal Print, **remove the old shared queue** (or block new connections) for that printer; leave it briefly disabled before deletion as a rollback hedge.

3. Once **all** migrated printers show no production traffic on the old server, decommission the print server (remove the print role / power down the box) and remove leftover GPO printer mappings.

4. Keep the connector host running — it remains the bridge for connector-attached printers.

Order of operations

`Assess → Enable licensing → Install connector + register → Share in Entra ID → Manual smoke test → Build groups → Deploy to pilot → Validate (on/off network) → Pilot gate → Expand → Retire servers`

5. Testing Guide — Universal Print Deployment & Print-Server Retirement

Run after the pilot deployment applies (Implementation Step 7) and again before retiring any server.

Test 1 — Printer registered and shared

Do: Universal Print portal → **Printers**; confirm each migrated printer is **Registered** and **Shared** with the name `UniversalPrint – <Model/Location> – Shared`. Expected: Every targeted printer appears as Shared with the agreed name.

Test 2 — Connector healthy

Do: Universal Print portal → **Connectors**; check `UniversalPrint – Connector – Site1` status; run `UPRT-Validation.ps1` on the host. Expected: Connector shows **Active/Healthy**; the host has outbound HTTPS to the service.

Test 3 — Printer deploys automatically to the pilot

Do: On a synced pilot device (member of `UniversalPrint – Printers – Pilot`), open **Settings** → **Printers & scanners**. Expected: The deployed printer appears **without** a manual add.

Test 4 — Test print on the corporate network

Do: From the pilot device on the LAN, print a test page to the deployed printer (or run `UPRT-Validation.ps1`). Expected: The job completes and paper is produced.

Test 5 — Test print off network

Do: Disconnect from the corporate LAN (VPN off / home / hotspot) and print the same test page. Expected: The job completes — cloud routing works off network (the connector host must be reachable for connector-attached printers).

Test 6 — Deployment coverage report

Do: Run `UPRT-Report.ps1`. Expected: A per-printer / per-group deployment inventory showing the pilot group covered.

Test 7 — Old-queue traffic falling (pre-retirement)

Do: On the old print server, review the print queue's job activity over the coexistence window. Expected: Production jobs on the old queue trend to near zero as Universal Print takes over.

Pass/fail gate

Gate	Requirement
Pilot pass	Tests 1–6 pass; pilot devices print on **and** off network through Universal Print
Retirement gate	Test 7 confirms near-zero old-queue traffic for a printer before its old queue/server is removed

6. Troubleshooting Guide — Universal Print Deployment & Print-Server Retirement

Use `UPRT-Validation.ps1` / `UPRT-Discovery.ps1` for device and print-environment state, `UPRT-Report.ps1` for deployment coverage, and `UPRT-Remediation.ps1` (default `-WhatIf`) for safe fixes.

A. Printer not appearing on a device

Symptom	Cause	Remediation
Deployed printer never shows up	Device not synced, or not in the target group	Sync the device; confirm membership of `UniversalPrint – Printers – Pilot/Broad`; run `UPRT-Remediation.ps1`
Some devices get it, others don't	Mixed group membership or stale assignment	Run `UPRT-Report.ps1` for coverage; reconcile group membership
Printer added manually but not via Intune	Printer shared but no Intune printer deployment	Create the Universal Print printer deployment and assign the group (Implementation Step 6)

B. Print jobs fail or hang

Symptom	Cause	Remediation
Jobs stuck "Sending"	Connector host offline/unreachable	Check `UniversalPrint – Connector – Site1` status in the portal; ensure the host is on and has outbound HTTPS
Works on LAN, fails off network	Connector host not reachable, or endpoint blocked off-site	Confirm the connector host stays reachable; allow Universal Print endpoints on the off-network path
Job errors immediately	Printer not Shared, or user lacks access	Confirm the printer is Shared; deploy via Intune group rather than per-user share

C. Connector problems

• Connector shows Offline: confirm the Windows service is running on the host, the host signed in with the Printer Administrator account, and outbound HTTPS to the service is open. Re-run `UPRT-Validation.ps1`.

• Printers missing from the connector: re-open the connector app and re-register the printers the host can see; large estates may need a second connector.

D. Licensing / volume

• Printer registration blocked or jobs rejected: confirm Universal Print licensing is assigned and print-volume allotment is not exhausted (Microsoft 365 admin center → Billing → Licenses).

E. Old queue and new printer both present (during coexistence)

• This is expected during coexistence. Once the printer's users are validated on Universal Print, remove the old shared queue (disable first) so users stop selecting the legacy path.

F. Diagnostics quick reference

```
Get-Printer | Select-Object Name, DriverName, PortName, Shared, Published
Get-PrintConfiguration -PrinterName "<name>"          # per-queue settings
Get-Service -Name "PrintWorkflowUserSvc*","Spooler"   # local print services
# Universal Print via Graph (uses Invoke-MgGraphRequest against /print endpoints):
Invoke-MgGraphRequest -Method GET -Uri "/v1.0/print/printers" | Out-String
Invoke-MgGraphRequest -Method GET -Uri "/v1.0/print/shares"   | Out-String
```

G. When to escalate

If the printer is Registered + Shared, the connector is healthy, the device is synced and in the correct group, endpoints are reachable on the failing network, and jobs still fail, collect the connector logs and `UPRT-Validation.ps1` output and open a Microsoft support case for Universal Print.

7. Best Practices — Universal Print Deployment & Print-Server Retirement

- **Pilot before you retire, always.** Prove printing on a small `UniversalPrint – Printers – Pilot` group, on and off network, before adding the broad group or touching a print server. The pilot gate is the single most important practice.
- **Coexist, then cut over.** Run the old queues and Universal Print side by side; only remove an old queue once that printer's users are fully on Universal Print and traffic has fallen to near zero.
- **Name everything consistently.** Use `<Purpose> – <Scope> – <Stage>` for the connector, shared printers, and deployment groups, and reuse the names verbatim. Consistent share names make deployment, reporting, and troubleshooting far easier.
- **Deploy by group, not by user.** Assign printers to Entra ID groups through Intune so new members get the printer automatically — assign once, repeat everywhere.
- **Mind the connector host.** It is the bridge for non-Universal-Print-ready printers; keep it always-on, patched, and reachable. Split large estates across multiple connectors to stay under printer-count and throughput ceilings.
- **Confirm off-network printing explicitly.** The whole point is print-from-anywhere; test from home/VPN-off, not just the LAN.
- **Disable before delete.** When retiring an old queue, disable it briefly before deletion so you have a fast rollback if a job path was missed.
- **Clean up the old plumbing.** Remove GPO printer mappings and leftover drivers after cutover so devices stop trying the old path.
- **Watch coverage and licensing.** Track deployment coverage with `UPRT-Report.ps1` and keep an eye on print-volume allotment as you expand.
- **Document the migration.** Record which printers are Universal Print-ready vs. connector-attached, their share names, group assignments, and the retirement date of each old server for audits and repeatability.

8. Deliverables Summary — Universal Print Deployment & Print-Server Retirement

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	What Universal Print is, the connector model, Entra ID sharing, Intune deployment, why retire the server
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing, plus the naming convention
Global Requirements (Infrastructure)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered steps (license, connector, share, deploy, validate, retire)
5. Testing Guide	Validation procedures with pass/fail and retirement gates
6. Troubleshooting Guide	Field issues and remediation
7. Best Practices	Field-proven recommendations

PowerShell scripts

Script	What it does
`UPRT-Discovery.ps1`	Discovers print servers/queues/drivers via the PrintManagement module (read-only)
`UPRT-Validation.ps1`	Universal Print readiness checks + a test print (read-only)
`UPRT-Report.ps1`	Printer + deployment inventory across queues and Universal Print (read-only)
`UPRT-Remediation.ps1`	Safely fixes failed printer deployments — default `-WhatIf`
`UPRT-Set-PrinterDeploymentTargeting.ps1`	Builds the printer-deployment groups and stages membership

What the buyer receives

- This implementation guide as a branded Word document and PDF.
- All five PowerShell scripts in a single ZIP.

• A repeatable process: Assess → Configure → Pilot → Validate → Expand, with a pilot gate before server retirement.

Outcome

Cloud-delivered printing for every user — printers shared from Entra ID and deployed through Intune, reachable on or off network — with the old print servers decommissioned, proven on a pilot before scale.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

UPRT-Discovery.ps1

Purpose: Inventories the current print environment using the PrintManagement module: installed printers (queues), their drivers, ports, and share/publish state, plus the installed driver list. Run on each print server so you know exactly what must be migrated to Universal Print and later retired. HOW TO RUN Elevated, on each print server: powershell -ExecutionPolicy Bypass -File .\UPRT-Discovery.ps1 Optional: -ComputerName to query a remote print server (needs RPC access). EXPECTED RESULTS A per-queue table (name, driver, port, shared/published) and the installed driver list, which becomes the registration plan for Universal Print. .NOTES Infrastructure solution - the repeatable way. Read-only (no changes made). Requires the PrintManagement module (in-box on Windows with the Print role / RSAT). #> [CmdletBinding()] param([string]$ComputerName = $env:COMPUTERNAME)

How to run: Elevated, on each print server: powershell -ExecutionPolicy Bypass -File .\UPRT-Discovery.ps1 Optional: -ComputerName <server> to query a remote print server (needs RPC access). EXPECTED RESULTS A per-queue table (name, driver, port, shared/published) and the installed driver list, which becomes the registration plan for Universal Print. .NOTES Infrastructure solution - the repeatable way. Read-only (no changes made). Requires the PrintManagement module (in-box on Windows with the Print role / RSAT). #> [CmdletBinding()] param([string]$ComputerName = $env:COMPUTERNAME)

UPRT-Remediation.ps1

Purpose: Reconciles printer-share deployment so a printer that should reach a group actually does: ensures the target Entra ID group is granted access to the printer share, and (on a device) forces an MDM/Intune sync so the deployed printer re-applies. Defaults to -WhatIf: it reports what it WOULD change and makes no modifications until you pass -WhatIf:$false. HOW TO RUN Preview (default, no changes): powershell -ExecutionPolicy Bypass -File .\UPRT-Remediation.ps1 ` -ShareName "UniversalPrint - HQ-Floor2-MFP - Shared" -GroupId Apply changes: ... -ShareName "..." -GroupId -WhatIf:$false EXPECTED RESULTS With -WhatIf (default): a plan of the access grant + sync that would occur. With -WhatIf:$false: the group is granted to the share via Graph and a device MDM sync is triggered. SCOPES Requires PrinterShare.ReadWrite.All (delegated) for the access grant. .NOTES Infrastructure solution - the repeatable way. Default -WhatIf; makes changes only when disabled. #> [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'High')] param([Parameter(Mandatory = $true)][string]$ShareName, [Parameter(Mandatory = $true)][string]$GroupId, [switch]$SyncDevice)

How to run: Preview (default, no changes): powershell -ExecutionPolicy Bypass -File .\UPRT-Remediation.ps1 ` -ShareName "UniversalPrint - HQ-Floor2-MFP - Shared" -GroupId <entra-group-guid> Apply changes: ... -ShareName "..." -GroupId <guid> -WhatIf:$false EXPECTED RESULTS With -WhatIf (default): a plan of the access grant + sync that would occur. With -WhatIf:$false: the group is granted to the share via Graph and a device MDM sync is triggered. SCOPES Requires PrinterShare.ReadWrite.All (delegated) for the access grant. .NOTES Infrastructure solution - the repeatable way. Default -WhatIf; makes changes only when disabled. #> [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'High')] param([Parameter(Mandatory = $true)][string]$ShareName, [Parameter(Mandatory = $true)][string]$GroupId, [switch]$SyncDevice)

UPRT-Report.ps1

Purpose: Produces a consolidated inventory across two surfaces: (1) local/legacy print queues via the PrintManagement module, and (2) Universal Print registered printers, shares, and printer-share deployments via Microsoft Graph. Use it to track migration coverage - which printers are registered/shared and which are deployed to groups - before retiring print servers. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\UPRT-Report.ps1 Requires the Microsoft Graph PowerShell SDK and consent for Printer.Read.All and PrinterShare.Read.All (delegated). Sign in when prompted. EXPECTED RESULTS A table of legacy queues, then a table of Universal Print printers/shares with their deployment targets (group IDs), so coverage gaps are obvious. .NOTES Infrastructure solution - the repeatable way. Read-only (GET only against Graph). #> [CmdletBinding()] param()

How to run: powershell -ExecutionPolicy Bypass -File .\UPRT-Report.ps1 Requires the Microsoft Graph PowerShell SDK and consent for Printer.Read.All and PrinterShare.Read.All (delegated). Sign in when prompted. EXPECTED RESULTS A table of legacy queues, then a table of Universal Print printers/shares with their deployment targets (group IDs), so coverage gaps are obvious. .NOTES Infrastructure solution - the repeatable way. Read-only (GET only against Graph). #> [CmdletBinding()] param()

UPRT-Set-PrinterDeploymentTargeting.ps1

Purpose: Creates the Entra ID security groups used to target Universal Print printer deployment, using the solution naming convention, and optionally stages initial members. These groups are then assigned to printer deployments in Intune (or granted to printer shares in Graph). Default object names: UniversalPrint - Printers - Pilot UniversalPrint - Printers - Broad HOW TO RUN Create both groups (idempotent - reuses any group that already exists): powershell -ExecutionPolicy Bypass -File .\UPRT-Set-PrinterDeploymentTargeting.ps1 Create and add pilot members: ... -PilotMemberIds @('','') Requires Group.ReadWrite.All (delegated). Sign in when prompted. EXPECTED RESULTS The two security groups exist (created or found) and any supplied pilot members are added; the script prints each group's object id for use in Intune assignment / Graph share grants. .NOTES Infrastructure solution - the repeatable way. Creates groups only; does not assign printers. #> [CmdletBinding(SupportsShouldProcess = $true)] param([string]$PilotGroupName = 'UniversalPrint - Printers - Pilot', [string]$BroadGroupName = 'UniversalPrint - Printers - Broad', [string[]]$PilotMemberIds = @())

How to run: Create both groups (idempotent - reuses any group that already exists): powershell -ExecutionPolicy Bypass -File .\UPRT-Set-PrinterDeploymentTargeting.ps1 Create and add pilot members: ... -PilotMemberIds @('<entra-object-id-1>','<entra-object-id-2>') Requires Group.ReadWrite.All (delegated). Sign in when prompted. EXPECTED RESULTS The two security groups exist (created or found) and any supplied pilot members are added; the script prints each group's object id for use in Intune assignment / Graph share grants. .NOTES Infrastructure solution - the repeatable way. Creates groups only; does not assign printers. #> [CmdletBinding(SupportsShouldProcess = $true)] param([string]$PilotGroupName = 'UniversalPrint - Printers - Pilot', [string]$BroadGroupName = 'UniversalPrint - Printers - Broad', [string[]]$PilotMemberIds = @())

UPRT-Validation.ps1

Purpose: Validates that this host (a device or the connector host) is ready for Universal Print: OS build, spooler service, outbound HTTPS reachability to the Universal Print endpoints, and that a named Universal Print printer is present locally. Optionally sends a test page to a deployed printer to prove end-to-end printing on the current network. HOW TO RUN powershell -ExecutionPolicy Bypass -File

.\UPRT-Validation.ps1 ` -PrinterName "UniversalPrint - HQ-Floor2-MFP - Shared" -TestPrint Omit -TestPrint for read-only readiness checks only. EXPECTED RESULTS PASS/WARN lines for OS, spooler, endpoint reachability, and printer presence; if -TestPrint is supplied and the printer is present, a test page is queued to it. .NOTES Infrastructure solution - the repeatable way. Read-only unless -TestPrint is specified. #> [CmdletBinding()] param([string]$PrinterName, [switch]$TestPrint)

How to run: powershell -ExecutionPolicy Bypass -File .\UPRT-Validation.ps1 ` -PrinterName "UniversalPrint - HQ-Floor2-MFP - Shared" -TestPrint Omit -TestPrint for read-only readiness checks only. EXPECTED RESULTS PASS/WARN lines for OS, spooler, endpoint reachability, and printer presence; if -TestPrint is supplied and the printer is present, a test page is queued to it. .NOTES Infrastructure solution - the repeatable way. Read-only unless -TestPrint is specified. #> [CmdletBinding()] param([string]$PrinterName, [switch]$TestPrint)